A Low-Resource Speech-to-Speech Translation System for Sylheti to Bangla and English

A thesis submitted in partial fulfilment of the requirements for the degree of

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Declaration

We hereby declare that this thesis, titled "A Low-Resource Speech-to-Speech Translation System for Sylheti to Bangla and English", is the result of our own work carried out under the supervision of [Supervisor name], [Designation], Department of [Department], [University].

All sources of information, data and software used in this work have been duly acknowledged and cited. In particular, the Sylheti sentences and their Standard Bangla and English translations are taken from the publicly available Kothon dataset [30], and the speech recordings were collected for this thesis with the informed consent of the speakers.

We further declare that this thesis, in whole or in part, has not been submitted to any other university or institution for the award of any degree or diploma.

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Abstract

Sylheti, a regional language of north-eastern Bangladesh and the Barak Valley of India, is spoken by millions of people but has almost no speech technology, and general-purpose models do not handle it. This thesis presents a cascaded speech-to-speech translation system from Sylheti to Standard Bangla and English. We record a speech corpus of the 9,320 Sylheti sentences of an existing text corpus, which provides Bangla and English translations, read by native speakers in two recording rounds (Round 1: 27,052 recordings, 30.13 hours, five speakers; Round 2: 10,629 recordings, 13.59 hours, 12 new speakers), and clean it with an automatic recording check that corrected 688 misplaced Round 1 recordings. Whisper-large-v3-turbo and Whisper-small are fine-tuned for Sylheti speech recognition on the same data. Fine-tuning reduces the word error rate (WER) of Whisper-large-v3-turbo from above 100% to 37.3% on seen speakers and 33.9% on a speaker held out from training, compared with 43.1% and 42.5% for Whisper-small. Adding nine of the Round 2 speakers to the training data, with the same settings, lowered the WER of Whisper-large-v3-turbo further, by 4.1 points on three new speakers held out from training and by 2.6 points on the held-out Round 1 speaker, both significant in a paired bootstrap test ($p \leq 0.001$). Whisper-small gained even more from the additional speakers (6.5 and 4.9 points) but remained less accurate than Whisper-large-v3-turbo. A single NLLB-200-distilled-600M model, fine-tuned for both directions, reaches 51.32 BLEU (Sylheti → Bangla) and 38.25 BLEU (Sylheti → English) on gold text. In the full cascade with Whisper-large-v3-turbo, ASR errors reduce translation quality by 14–17 BLEU for Bangla and 7–9 BLEU for English, which identifies speech recognition as the main bottleneck. With the Round 2 ASR models, translation quality rose significantly for the held-out Round 1 speaker and the Round 2 speakers (up to +6.6 BLEU for Bangla), while the gain on the three new speakers was too small to be confirmed. Bangla and English speech is generated with MMS-TTS. We show that automatic evaluation of Bangla TTS by Whisper back-transcription is unreliable: Whisper mixes Devanagari letters into most Bangla transcripts and inserts extra words. In an additional experiment, fine-tuning MMS-TTS on 1.9 hours of one speaker's Sylheti speech, with the pronunciation and timing modules kept fixed, produced speech in that speaker's voice with 28.1% CER, compared with 16.8% for the original voice.

Keywords: Sylheti, low-resource languages, speech recognition, machine translation, speech-to-speech translation, Whisper, NLLB-200

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[VERIFY: to be written by the authors — supervisor, the speakers who contributed recordings (without names if they prefer), and others who helped.]

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List of Abbreviations

Abbreviations used in this thesis.

Abbreviation	Meaning
ASR	Automatic Speech Recognition
BLEU	Bilingual Evaluation Understudy
CER	Character Error Rate
chrF	Character n-gram F-score
CI	Confidence interval
LR	Learning rate
MMS	Massively Multilingual Speech
MOS	Mean Opinion Score
MT	Machine Translation
NLLB	No Language Left Behind
RTF	Real-Time Factor
S2ST	Speech-to-Speech Translation
TTS	Text-to-Speech
WER	Word Error Rate

Chapter 1: Introduction

1.1 Background and Motivation

Sylheti (ISO 639-3: syl) is spoken in the Sylhet Division of north-eastern Bangladesh, in the Barak Valley of Assam and in northern Tripura in India, and by a large diaspora, especially in the United Kingdom. It is spoken by an estimated 11.8 million people according to the 22nd edition of Ethnologue [1]. Sylheti is not officially recognised as a language in either Bangladesh or India and is variously treated as a dialect of Bengali or as a language of its own [1], [2]. It has no standardised writing system: the historical Sylheti Nagri script is little used today, and most speakers who write Sylheti use the Bengali–Assamese script.

Although many Sylheti speakers also understand Standard Bangla, the two differ substantially in pronunciation and vocabulary. As a result, everyday Sylheti speech is poorly served by language technology built for Standard Bangla or English.

Speech-to-speech translation (S2ST) would let a Sylheti speaker be understood by listeners who know only Standard Bangla or English, for example in health care, public services or communication within the diaspora. Recent multilingual models for speech recognition, translation and speech synthesis make such systems feasible for languages with little data, provided they can be adapted with a modest amount of in-language material.

1.2 Problem Statement

General-purpose models do not handle Sylheti: the untrained Whisper-large-v3-turbo model produces a word error rate above 100% on our Sylheti test sets, and untrained NLLB-200 translates Sylheti text to Bangla worse than simply copying the input. The problem addressed in this thesis is to build and evaluate a system that takes Sylheti speech and produces understandable Bangla and English speech, using a self-collected corpus of modest size.

1.3 Research Gap in Brief

To the best of our knowledge, existing speech resources for regional Bangla include Sylheti only as one of several regional varieties, mostly without translations, and they target recognition or conversion to Standard Bangla rather than a complete speech-to-speech system. We are not aware of a Sylheti speech corpus of this size aligned

with both Standard Bangla and English, of a Sylheti → Bangla / English S2ST system evaluated module by module and end to end, or of an evaluation on speakers unseen in training. Section 2.7 states the gap in detail.

1.4 Objectives and Research Questions

The objectives of this thesis are to:

1. Build and clean a Sylheti speech corpus aligned with Bangla and English text.
2. Fine-tune ASR models for Sylheti and compare a large model with a small baseline.
3. Fine-tune one MT model for Sylheti → Bangla and Sylheti → English.
4. Generate Bangla and English speech with an open multilingual TTS model.
5. Evaluate each module and the full cascade, including on a speaker unseen in training.
6. Measure how adding speakers recorded with a web-based tool changes ASR accuracy, especially for speakers unseen in training.

The research questions are:

- **RQ1:** How much does fine-tuning improve Whisper on Sylheti, and does a larger model generalise better to a new speaker?
- **RQ2:** How much translation quality is lost because of ASR errors in the cascade?
- **RQ3:** How reliable is ASR back-transcription for measuring the intelligibility of Bangla TTS?
- **RQ4:** How do the results change when more speakers are added?

1.5 Contributions

1. A Sylheti speech corpus of 9,320 sentences with Bangla and English translations (37,681 recordings, 43.72 hours from 17 speakers in two recording rounds), together with an automatic recording check that corrected 688 misplaced Round 1 recordings and restored 9,266.
2. A Sylheti → Bangla / English speech-to-speech translation pipeline built from fine-tuned Whisper-large-v3-turbo, fine-tuned NLLB-200-distilled-600M and pretrained TTS models.
3. A controlled comparison of Whisper-large-v3-turbo and Whisper-small on the same data, with per-speaker and unseen-speaker evaluation.
4. A measurement of how ASR errors reduce translation quality in the cascade, using gold Sylheti text as an upper bound.

5. Evidence that Whisper back-transcription misjudges Bangla TTS intelligibility (Devanagari output and inserted words).
6. A second recording round with a web-based recorder (12 new speakers, 10,629 recordings) and a controlled comparison of ASR trained with and without these speakers, with significance tests on three additional unseen speakers.

1.6 Thesis Organisation

Chapter 2 reviews related work and states the research gap. Chapter 3 describes the dataset and the recording check. Chapter 4 presents the methodology, and Chapter 5 the results, including the effect of adding the Round 2 speakers (Section 5.8). Chapter 6 discusses the findings and limitations, and Chapter 7 concludes and outlines future work.

Chapter 2: Literature Review

This chapter reviews the models and resources this thesis builds on: speech recognition, translation and speech synthesis for low-resource languages, speech-to-speech translation, and existing speech data for regional varieties of Bangla. It ends with the research gap addressed in this thesis.

2.1 Low-Resource Automatic Speech Recognition

Whisper [3] is trained with weak supervision on a very large multilingual collection of audio paired with transcripts and can transcribe many languages without fine-tuning. Whisper-large-v3-turbo [4], used as the main ASR model in this thesis, is a released variant of the Whisper large model with a reduced decoder for faster inference; Whisper-small is a smaller member of the same family. Massively Multilingual Speech (MMS) [5] extends speech recognition, language identification and speech synthesis to more than a thousand languages. Because Sylheti is not among Whisper's training languages and its spelling is not standardised, fine-tuning on in-language data is required (Chapter 5). Data augmentation such as SpecAugment [6], which masks blocks of time and frequency in the input features, is commonly used to reduce overfitting when the fine-tuning set is small.

2.2 Speech Data for Regional Bangla

RegSpeech12 [7] is a corpus of spontaneous Bengali speech from several regions of Bangladesh, including Sylhet, released to study speech recognition for regional varieties. Rahman et al. [8] provide a dataset of 5,200 utterances from eight districts of Bangladesh (650 per district, including Sylhet) for translating regional Bangla speech into Standard Bangla and English. Both resources were examined during this work but not used in the reported experiments. They treat Sylheti as one of many regional varieties, with comparatively little Sylheti material, and are aimed at recognition or conversion to Standard Bangla rather than a complete Sylheti speech-to-speech system. Ben-10 [25] contains about 78 hours of mostly spontaneous speech from ten regions including Sylhet; benchmarks on it show that Whisper-large-v3 reaches a WER above 100% on regional Bangla, and that fine-tuning is needed. BanglaTalk [26] builds a real-time spoken assistant for regional dialects with a fine-tuned wav2vec 2.0 model, a language model and VITS or MMS speech synthesis, and BanglaDialecto [27] transcribes Noakhali speech with a

fine-tuned Whisper model and standardises the text with BanglaT5. Earlier Sylheti ASR used small-vocabulary convolutional network models [35], and a Sylheti TTS system for the Sylheti Nagri script was trained with VITS on 15 hours of one voice [34].

2.3 Machine Translation for Low-Resource Languages

No Language Left Behind (NLLB-200) [9] is a multilingual translation model covering 200 languages, including Bengali, and is the MT model used in this thesis. NLLB-200 has no language code for Sylheti; since Sylheti is written here in the Bengali script, the Bengali code (ben_Beng) is used for the Sylheti source side, and the model is fine-tuned so that it learns Sylheti input. Several text corpora pair regional Bangla with Standard Bangla: Vashantor [28] covers five regions including Sylhet, ONUBAD [29] covers Chittagong, Sylhet and Barisal, and Kothon [30] provides about 9,300 Sylheti and 8,000 Chittagonian sentences. Sylheti MT has been studied with LSTM and sequence-to-sequence models [33], with fine-tuned mBART-50 and MarianMT for Bengali → Sylheti [32], and with prompted large language models in both directions between Sylheti and Standard Bangla [31]. These studies translate between Sylheti and Standard Bangla only.

2.4 Text-to-Speech for Bangla

MMS [5] provides VITS-based [10] TTS models for over a thousand languages, including Bengali, and is the main TTS model in this thesis. Parler-TTS [11] generates speech controlled by a natural-language description of the voice, following Lyth and King [12]. Indic Parler-TTS [13], [14] is a version trained for Indian languages, including Bengali. Intelligibility of TTS is often estimated automatically by transcribing the generated speech with an ASR model and computing error rates; this thesis shows that this method is unreliable for Bangla when the evaluating model writes parts of the output in another script (Section 5.6).

2.5 Speech-to-Speech Translation

Speech-to-speech translation can be built as a cascade of ASR, MT and TTS, or as a direct (end-to-end) model that maps source speech to target speech. Direct and multilingual end-to-end models need large amounts of parallel speech or strong pretraining in the source language. For a language without parallel speech data and without a writing standard, a cascade is the practical choice, and it allows each stage to be evaluated separately.

2.6 Evaluation Metrics

ASR is evaluated with word error rate (WER) and character error rate (CER). Translation is evaluated with BLEU [15], based on word n-gram overlap, and chrF [16], based on character n-grams, which is better suited to morphologically rich languages and spelling variation. SacreBLEU [17] provides standardised implementations of both so that scores are comparable across studies.

2.7 Research Gap

Recent work has produced several resources for regional varieties of Bangla, but, to the best of our knowledge, they leave four gaps for Sylheti speech translation.

- Corpus. Regional speech corpora such as Ben-10 [25] and RegSpeech12 [7] include Sylhet among ten to twelve regions and provide transcribed, mostly spontaneous speech without translations, while text corpora such as Vashantor [28], ONUBAD [29] and Kothon [30] pair Sylheti with Standard Bangla but contain no speech. No corpus combines read Sylheti speech from many speakers with aligned Standard Bangla and English text. This thesis records the 9,320 sentences of [30] with native speakers: 27,052 recordings (30.1 h) in Round 1 and 10,629 recordings (13.6 h) in Round 2 (Chapter 3).
- Speech recognition. Foundation models such as Whisper fail on regional Bangla without adaptation (WER above 100% on Ben-10 [25]), and earlier Sylheti ASR used small-vocabulary models [35]. Model size, generalisation to speakers absent from training and the effect of adding speakers have not been studied for Sylheti. This thesis compares Whisper-large-v3-turbo and Whisper-small on identical data, evaluates both on held-out speakers, and tests the effect of nine additional speakers with a paired bootstrap (Sections 5.1–5.3 and 5.8).
- Translation and the full system. Sylheti MT has so far translated only between Sylheti and Standard Bangla, with LSTM models [33], mBART-50 and MarianMT [32] or prompted large language models [31]; translation from Sylheti into English has not been reported. Existing regional speech pipelines either standardise another dialect into text (BanglaDialecto, Noakhali [27]) or answer spoken questions rather than translate (BanglaTalk [26]). No Sylheti → Bangla / English speech-to-speech system has been evaluated module by module, and the loss of translation quality caused by ASR errors

is unknown. This thesis fine-tunes one NLLB-200 model for both target languages and measures the cascade against a gold-text upper bound (Sections 5.4–5.5).

- Speech synthesis. Sylheti TTS has been built for the Sylheti Nagri script from 15 hours of one professional voice [34]. The reliability of automatic Bangla TTS evaluation by ASR back-transcription, and adapting a multilingual TTS model to a Sylheti speaker in Bengali script from under two hours of speech, have not been examined. This thesis shows that Whisper back-transcription is unreliable for Bangla because of script mixing and inserted words, and adapts MMS-TTS to one Sylheti speaker (Section 5.6).

Chapters 3–5 address these gaps in turn.

Chapter 3: Dataset

The corpus was recorded in two rounds. Round 1 contains 27,052 recordings (30.13 hours) from five speakers (A1, A2, A4, A6, A7); Round 2 adds 10,629 recordings (13.59 hours) from 12 new speakers (B01–B12), collected with a web-based recorder. Both rounds use the same 9,320 Sylheti sentences with their Bangla and English translations. This chapter describes the corpus, the recording procedure, the automatic recording check, the Round 1 split (Section 3.6) and the Round 2 data (Section 3.7).

3.1 Corpus Design

The corpus is built around 9,320 Sylheti sentences. Each sentence has a Standard Bangla translation and an English translation, so the same text unit can be used to train and evaluate automatic speech recognition (ASR), machine translation (MT) and the full speech-to-speech pipeline. The sentences and their translations are taken from the Kothon dataset [30], in which Standard Bangla sentences collected from open-source resources, novels, existing datasets and printed books were translated into Sylheti by native-speaker annotators and checked by independent native speakers; the English column is a machine translation of the Standard Bangla sentence. This thesis uses only the Sylheti part of Kothon and adds speech recordings of its sentences.

3.2 Recording Procedure

The sentences were read aloud by native Sylheti speakers, giving a read-speech corpus. Recordings were made with [device: e.g. the speakers' own smartphones] in [environment: e.g. quiet indoor rooms]. Speakers were asked to read each sentence naturally at a normal pace and to re-record any sentence containing a reading mistake or background noise. Round 1 recordings were collected between [month year] and [month year]. Each speaker is identified only by a code (A1, A2, A4, A6, A7).

To collect further speakers for Round 2, a web-based recorder was developed. A volunteer opens a link on a phone, enters a speaker code, gives consent and records the sentence list one sentence at a time; each recording is stored with its speaker code and sentence number. Recording starts only after the volunteer has agreed to the consent statement shown by the recorder. In Round 2, 12 new volunteers (B01–B12) recorded with this tool (Section 3.7).

3.3 Ethics and Consent

All speakers are anonymised by code, and no names are reported in this thesis or released with the data. Recordings from speakers who withdrew consent were removed from the dataset and are not used in any experiment. All speakers, including the Round 2 volunteers (B01–B12), gave written informed consent by signing a consent form after being told that their recordings would be used for research and released only under a speaker code. [Ethics: formal ethics approval was not required by the department / was obtained from (committee).]

3.4 Pre-processing

All recordings were converted to 16 kHz mono WAV, the input format expected by the ASR models, and leading and trailing silence was trimmed. Clips longer than 30 seconds are excluded from ASR training and testing. Two files of speaker A6 that are listed in the metadata are missing from the audio archive; they are skipped, so 20,920 of the 20,922 training recordings are used.

3.5 Automatic Recording Check (Round 1)

When many people read a long sentence list, some recordings end up attached to the wrong sentence: a speaker skips a line, repeats one or reads ahead. Checking 36,582 recordings by ear was not feasible, so an automatic check was applied to every recording (Step 1b, refined in Step 1c).

3.5.1 Transcription

Each recording was transcribed with the untrained Whisper-large-v3-turbo model (fp16, language set to Bengali). Audio shorter than 0.1 s was treated as empty. Whisper has not seen Sylheti, so its transcripts are noisy, but they are usually close enough in sound to the intended sentence to identify it.

3.5.2 Fuzzy Matching

Each transcript was compared with all 9,320 sentences, not only with the sentence it was filed under. Before comparison, the transcript and both the Sylheti and Bangla text of every sentence were converted to a sound-folded key: letters that are pronounced alike are merged (for example খ→ক, ছ→চ, ফ→প, ভ→ব, শ/ষ→স, long vowels → short vowels), and diacritics such as hasanta and chandrabindu are removed. When Whisper wrote the audio in another script (Latin, Devanagari or Arabic letters), a script-independent

consonant skeleton was used instead. Similarity was measured with the RapidFuzz ratio [18] on a 0–100 scale; *score_filed* is the similarity to the filed sentence, and *score_best* and *score_second* are the two highest similarities over the whole corpus.

3.5.3 Verdicts

Each recording received exactly one verdict, decided by the thresholds in Table 3.2:

- **ok** — the filed sentence matches well (score_filed ≥ 55), or nearly as well as the best sentence (within 6 points), or the best sentence is textually identical to the filed one.
- **moved** — not ok, and another sentence matches clearly better: score_best ≥ 55, at least 15 points above score_filed, and at least 3 points above the second-best sentence.
- **shift** — at least three consecutive recordings of the same speaker are moved by the same offset (for example, all one sentence ahead).
- **moved_clash / shift_clash** — a moved or shifted recording whose target sentence already has an ok recording from the same speaker, so the recording is probably a duplicate take and is not moved.
- **check_ear** — the transcript is too short (sound-folded key under 4 characters) or matches nothing (score_best < 25), so the recording can only be judged by listening.
- **uncertain** — none of the above: neither the filed sentence nor any other sentence matches clearly.

Table 3.1: Recording-check verdicts per speaker (Step 1b/1c).

Speaker	ok	uncertain	check_ear	moved	shift	moved_clash	Total
A1	825	81	79	10	6	0	1,001
A2	8,061	727	506	23	0	1	9,318
A4	4,654	2,706	882	898	33	20	9,193
A6	6,968	1,536	318	489	3	19	9,333
A7	6,263	1,057	335	62	15	5	7,737
Total	**26,771**	**6,107**	**2,120**	**1,482**	**57**	**45**	**36,582**

Table 3.2: Matching thresholds of the recording check (similarity 0–100).

Threshold	Value	Used for
OK_SCORE	55	filed sentence matches well
CLOSE	6	filed sentence nearly as good as the best
MOVE_MIN / MOVE_MARGIN	55 / 15	another sentence clearly better
POOR	25	nothing matches (check_ear)
RUN_MIN	3	consecutive moves form a shift
STRONG_SCORE / STRONG_MARGIN	65 / 20	a single move is safe to apply automatically

3.5.4 Corrections

All shifted recordings, and moved recordings with score_best ≥ 65 and at least 20 points above score_filed, were moved automatically to the matching sentence, unless this caused a clash or two recordings competed for the same target; in total 688 misplaced recordings were relabelled. Moved or shifted recordings that could not be relabelled safely were left out. An earlier run of the check also excluded all check_ear recordings; because these are mostly recordings that Whisper simply failed to transcribe, Step 1c kept them, together with all ok and uncertain recordings, which restored 9,266 recordings to the dataset. Speaker A4 needed the most corrections (898 moved recordings), which is consistent with A4 also being the hardest speaker for ASR (Chapter 5).

3.6 Data Split and Statistics (Round 1)

The data were split 80/10/10 by sentence with a fixed seed (13), so that no test sentence appears in training under any speaker. Speaker A2 was held out completely and is used only for the test sentences, giving an unseen-speaker test set.

Table 3.3: Round 1 data split.

Split	Speakers	Recordings	Hours
train	A1, A4, A6, A7	20,922	23.43
validation	A1, A4, A6, A7	2,605	2.91
test_seen	A1, A4, A6, A7	2,597	2.94
test_unseen	A2	928	0.85
Total	5 speakers	**27,052**	**30.13**

Table 3.4: Recordings per speaker and split.

Speaker	train	validation	test_seen	test_unseen	Total
A1	800	102	91	–	993
A2	–	–	–	928	928
A4	6,761	841	838	–	8,440
A6	7,208	900	903	–	9,011
A7	6,153	762	765	–	7,680

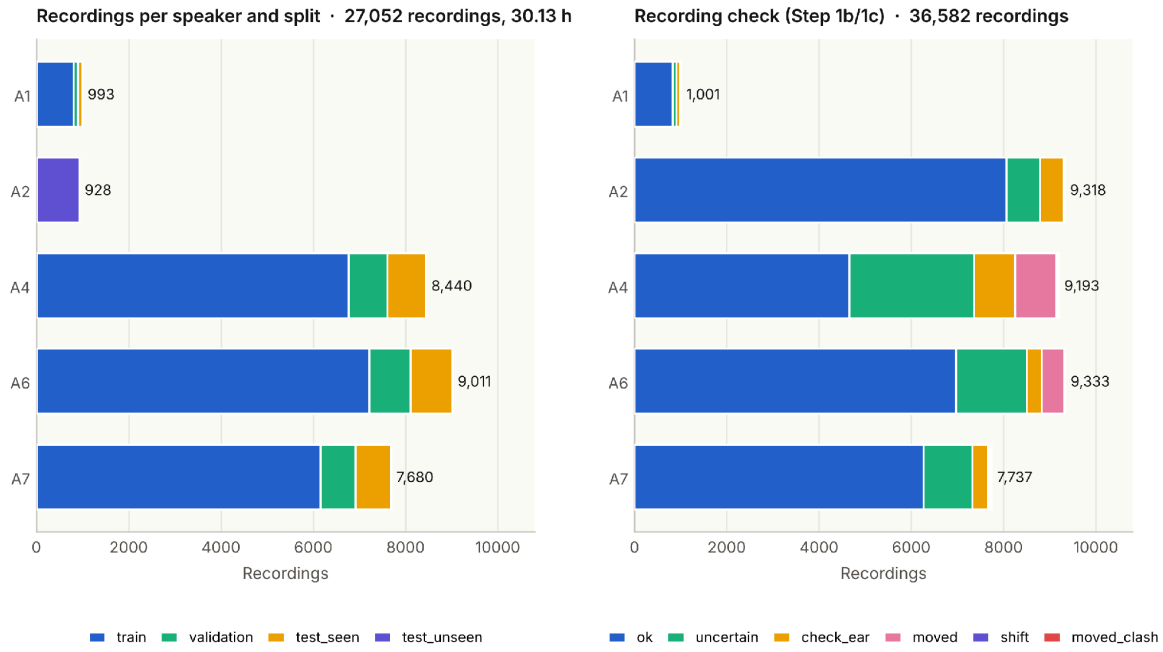


Figure 3.1: Recordings per speaker and split (left) and recording-check verdicts per speaker (right).

The split is strongly unbalanced across speakers: A1 contributes only 800 training recordings, while A4, A6 and A7 each contribute more than 6,000. For MT, which uses text only, the same sentence-level split gives 7,456 training, 932 validation and 932 test sentences.

3.7 Round 2 Data

In Round 2, 12 new volunteers (B01–B12) recorded the sentence list with the web-based recorder (Section 3.2), which stores each sentence as a separate file. The recordings were converted to 16 kHz mono WAV and trimmed of leading and trailing silence as in Round 1 (Section 3.4). Of 10,693 recordings, 54 could not be decoded and were excluded, leaving 10,639 recordings (13.61 hours); two further recordings were shorter than 1 s or longer than 30 s. Because each sentence is stored as its own file, misplaced recordings are much less likely than in Round 1, and a simpler check was used: each recording was transcribed with the fine-tuned Round 1 ASR model (Section 5.2) and compared with its sentence by CER. 10,508 recordings were ok (CER $\leq$ 40%), 120 uncertain (CER $\leq$ 80%, kept as in Round 1), 4 matched another sentence better and 5 were removed (Table 3.5). Where a speaker recorded a sentence twice, the recording with the lower CER was kept. In total, 10,629 recordings (13.59 hours) were kept. Unlike Round 1, where the untrained Whisper model was used (Section 3.5), this check used the fine-tuned model, so the two rounds were not filtered in exactly the same way.

Table 3.5: Round 2 recording check (Step 9), all twelve speakers. Moved recordings were relabelled and kept; of two takes of the same sentence, the one with the lower CER was kept. The last column splits the kept recordings by the Round 1 sentence split (test / train / validation).

Speaker	Checked	ok	uncertain	moved	removed	duplicate takes	kept	hours	test / train / val
B01	1,110	1,100	10	0	0	0	1,110	1.41	94 / 904 / 112
B02	441	439	2	0	0	0	441	0.57	35 / 359 / 47
B03	1,034	1,028	5	0	1	0	1,033	1.29	109 / 824 / 100
B04	508	505	2	1	0	0	508	0.59	53 / 403 / 52
B05	1,225	1,190	35	0	0	0	1,225	1.57	131 / 977 / 117
B06	201	199	1	1	0	0	201	0.37	19 / 154 / 28
B07	1,318	1,304	12	0	2	3	1,313	1.59	119 / 1,068 / 126
B08	1,921	1,920	1	0	0	0	1,921	2.27	184 / 1,545 / 192
B09	500	497	3	0	0	0	500	0.57	40 / 407 / 53
B10	1,071	1,030	39	0	2	0	1,069	1.61	118 / 848 / 103
B11	1,299	1,287	10	2	0	0	1,299	1.73	120 / 1,054 / 125
B12	9	9	0	0	0	0	9	0.02	1 / 7 / 1
Total	**10,637**	**10,508**	**120**	**4**	**5**	**3**	**10,629**	**13.59**	**1,023 / 8,550 / 1,056**

Round 2 recordings follow the sentence-level split of Round 1 (Section 3.6), so no test sentence of either round appears in training. Speakers B02, B04 and B09 were held out completely as new unseen speakers. For the other nine speakers, only recordings of training sentences are added to the training set, and their recordings of test sentences form an additional test set (test_r2_seen). Validation uses the same 300 Round 1 clips in both rounds, so validation scores are directly comparable. Table 3.6 lists the sets used in the Round 2 experiment (Section 5.8).

Table 3.6: Training, validation and test sets in the Round 2 experiment. Round 1 hours include the two missing A6 files; hours of the 300-clip validation subset were not computed.

Set	Speakers	Recordings	Hours	Use
train, Round 1	A1, A4, A6, A7	20,920	23.43	both models
train, Round 2	B01, B03, B05–B08, B10–B12	7,381	9.54	Round 2 model only
train, total	13 speakers	28,301	32.97	Round 2 model
validation	A1, A4, A6, A7	300	–	checkpoint selection
test_seen	A1, A4, A6, A7	2,597	2.94	Round 1 training speakers
test_unseen	A2	928	0.85	unseen speaker
test_new	B02, B04, B09	128	0.15	new unseen speakers, test sentences
test_new_all	B02, B04, B09	1,449	1.73	new unseen speakers, all sentences
test_r2_seen	B01, B03, B05–B08, B10–B12	895	1.16	Round 2 training speakers, test sentences

Chapter 4: Methodology

The system is a cascade of three modules: ASR converts Sylheti speech to Sylheti text, MT translates the text into Bangla and English, and TTS turns the translations into speech. Each module is built from a pretrained model, adapted where Sylheti data is needed, and evaluated both on its own and inside the full pipeline.

4.1 System Overview

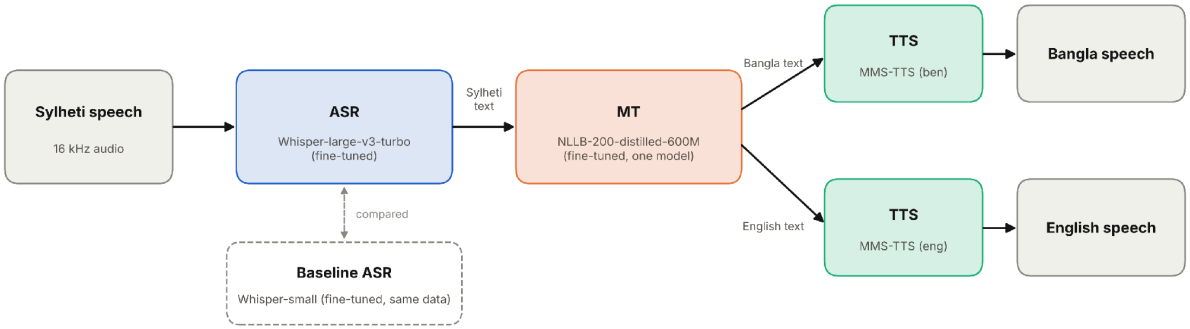


Figure 4.1: Cascaded speech-to-speech translation pipeline, with Whisper-small as the baseline ASR.

A cascade was chosen because no parallel Sylheti–Bangla or Sylheti–English speech data exists, while the corpus in Chapter 3 provides speech–text pairs for ASR and text–text pairs for MT. The cascade also makes each stage measurable on its own, which allows the effect of ASR errors on translation to be quantified (Section 4.5.3). All models are run with the Hugging Face Transformers library [19].

4.2 ASR Module

4.2.1 Models

The main ASR model is Whisper-large-v3-turbo (809M parameters), fully fine-tuned on the Sylheti training set. Whisper-small (244M parameters) is fine-tuned on exactly the same data and split as a smaller baseline. Both start from the public multilingual checkpoints, with the tokenizer and generation settings fixed to Bengali transcription; neither has seen Sylheti before fine-tuning.

4.2.2 Fine-tuning Configuration

Table 4.1: ASR fine-tuning configuration.

Setting	Whisper-large-v3-turbo	Whisper-small
Parameters	809M	244M

Setting	Whisper-large-v3-turbo	Whisper-small
Learning rate	1e-5, warm-up 300 steps, linear decay	1e-5, warm-up 300 steps, linear decay
Batch	8 × 4 (effective 32)	16 × 2 (effective 32)
Maximum length	5 epochs = 3,265 steps in Round 1; 4,420 steps in Round 2	5 epochs = 3,270 steps (654 steps/epoch) in Round 1; 4,420 steps in Round 2
Precision	bf16	fp16 (Round 1); bf16 (Round 2)
Optimiser	8-bit AdamW [20]	8-bit AdamW [20]
Memory	gradient checkpointing	gradient checkpointing
Augmentation	SpecAugment, mask_time_prob 0.05	SpecAugment, mask_time_prob 0.05
Evaluation / checkpoints	every 250 steps, 300 validation clips	every 250 steps, 300 validation clips
Label length	max. 448 tokens; clips ≤ 30 s	max. 448 tokens; clips ≤ 30 s
GPU	Colab L4 (Round 1); A100 (Round 2)	Colab T4 (Round 1); A100 (Round 2)

The two models differ only in settings forced by the hardware: the T4 GPU does not support bf16, so Whisper-small used fp16 and a per-device batch of 16 with 2 accumulation steps, keeping the same effective batch of 32. Other training arguments were left at the library defaults (weight decay 0, gradient clipping at norm 1.0). In Round 2, both models were trained on an A100 GPU with bf16.

Whisper-small was resumed after step 2,250 with a restarted learning-rate schedule: the saved training log shows a second 300-step warm-up from step 2,250 to 1e-5 at step 2,550, followed by linear decay (8.5×10^{-6} at step 3,000). Whisper-large-v3-turbo followed a single uninterrupted schedule, including when it was resumed from checkpoint 2,000 (3.4×10^{-6} at step 2,250, decaying to nearly zero at step 3,250). This difference is discussed in Chapter 6.

4.2.3 Model Selection

Both models were evaluated every 250 steps on a fixed subset of 300 validation clips, and a checkpoint was saved at each evaluation. Training stopped when the validation WER had not improved for 1,000 steps, and the checkpoint with the lowest validation WER was kept. Selection by WER rather than validation loss matters here: for Whisper-large-v3-turbo, validation loss still fell at step 1,250 while WER rose.

4.2.4 Decoding

Test transcriptions were generated with greedy decoding (no beam search) and a maximum of 440 new tokens, with the language set to Bengali and the task to transcription. During training, validation transcriptions were limited to 225 tokens, so validation and test scores are not computed under identical decoding limits.

4.3 MT Module

4.3.1 Model

MT uses NLLB-200-distilled-600M. A single model is fine-tuned for both directions, Sylheti → Bangla (target code ben_Beng) and Sylheti → English (eng_Latn), on the sentence-level text split (7,456 / 932 / 932 sentences). The Sylheti source side uses the Bengali code ben_Beng, as NLLB-200 has no Sylheti code. Because MT is trained on text only, it does not depend on the speech recordings and does not need retraining when speakers are added in Round 2.

4.3.2 Training and Inference Configuration

Table 4.2: MT fine-tuning and inference configuration.

Setting	Value
Model	NLLB-200-distilled-600M, one model, two directions
Language codes	source ben_Beng (Sylheti); targets ben_Beng, eng_Latn
Optimiser and learning rate	Adafactor [21], 1e-4
Maximum length	8 epochs (3,728 steps)
Evaluation	every 250 steps
Model selection	lowest validation loss (best: step 1,250, loss 0.9284); early stop at step 2,000
Inference (cascade)	beam search with 4 beams, batch 32, input truncated to 256 tokens, max. new tokens = min(256, 2 × input length + 10), fp16

Before translation, both ASR output and gold Sylheti text are cleaned with the same function used in MT training (Unicode NFC, punctuation replaced by spaces, repeated spaces collapsed).

4.4 TTS Module and Bangla Text Normalisation

Speech is generated with the pretrained MMS-TTS models (mms-tts-ben for Bangla, mms-tts-eng for English), without further training.

Before synthesis, Bangla text is normalised: Bangla and Latin digits are converted to Bangla number words (including thousands, lakhs, crores and decimals), symbols such as % and & are spelled out, quotation marks and brackets are removed, slashes are read as "অথবা" ("or"), repeated punctuation is collapsed, and each sentence ends with a daṇḍa (।). Sentences containing Latin letters are flagged, because the Bangla model cannot read them (Appendix E).

4.4.1 Speaker Adaptation (Voice Cloning)

As an additional experiment, the Bangla MMS-TTS model was adapted to the voice of one native Sylheti speaker, so that the system can speak its output in that voice. Speaker B01 recorded 1,110 sentences (5002–6761) with the web recorder; together with 482 Round 1 recordings of sentences 5501–6000 that were also read by B01 but are filed under speaker A4 in Round 1, this gives 1,592 recordings (about 1.9 hours) of one voice, referred to as V1 in this experiment. Each recording was transcribed with the fine-tuned Whisper-large-v3-turbo model and compared with its sentence; recordings with a CER above 50% or outside 1–20 s were removed, leaving 1,579. The web recorder's file numbers were found to be one higher than the corpus row and were corrected automatically.

The VITS model [10] was fine-tuned with the finetune-hf-vits recipe [22], which trains the generator together with a discriminator converted from the original MMS checkpoint for Bengali [5]. Three settings were compared (Table 5.8): fine-tuning the whole model (attempts 1 and 2), and fine-tuning only the posterior encoder, flow and decoder while keeping the text encoder and duration predictor fixed (attempt 3), so that pronunciation and timing stay as in the original model. All attempts were evaluated on the same 50 held-out sentences (web-recorder recordings of test-split sentences, never used in training): the generated speech was back-transcribed with the fine-tuned Whisper-large-v3-turbo model and scored with CER, using the default inference noise and noise_scale = noise_scale_duration = 0.3. Because this evaluator was fine-tuned on Sylheti, it avoids the Devanagari problem of the untrained Whisper model (Section 5.6). The notebooks are Step6_TTS_VoiceClone, Step6b_TTS_VoiceClone_v2 and Step6c_TTS_VoiceClone_v3.

4.5 Evaluation Metrics

4.5.1 WER and CER

ASR output is scored with WER and CER using the jiwer library [23]. Reference and hypothesis are normalised in the same way: Unicode NFC; zero-width characters removed (zero-width joiner kept only before র-phalā); punctuation (quotation marks, brackets, , . । ? ! : ; - / * … [ ] |) replaced by spaces; repeated spaces collapsed. No lower-casing is applied, as Bengali script has no case. An empty hypothesis is scored as “-”. WER above 100% is

possible when the output contains many insertions. Because Sylheti has no standard spelling, a correct word written in a different spelling is counted as an error.

4.5.2 BLEU and chrF

Translations are scored with corpus-level BLEU [15] and chrF [16] using sacreBLEU 2.4.3 [17]. chrF works on character n-grams and is less sensitive to small spelling differences than BLEU.

4.5.3 Cascade Evaluation and Upper Bound

In the cascade, the ASR output for each test recording is translated by the fine-tuned MT model and scored against the reference translation. The same recordings are also translated from their gold Sylheti text. The gold-text score is an upper bound for the cascade, and the difference between the two measures how much translation quality is lost through ASR errors.

4.5.4 TTS Intelligibility and Speed

TTS is evaluated on 100 MT translations sampled from the test set. Intelligibility is estimated by back-transcription: the generated audio is transcribed with Whisper-large-v3-turbo (language forced to Bengali or English) and compared with the text that was synthesised, using CER and WER. Because Whisper writes parts of Bangla transcripts in Devanagari letters, Bangla scores are also reported after mapping Devanagari characters to the corresponding Bengali characters. Speed is reported as real-time factor (RTF): generation time divided by audio duration.

4.5.5 Statistical Significance

Differences between two ASR models on the same test set are tested with a paired bootstrap over recordings [24]. The recordings of the test set are resampled with replacement 10,000 times; on each resample both models are scored on the same recordings, and the difference in corpus-level WER and CER is computed. The 2.5th and 97.5th percentiles of these differences give a 95% confidence interval, and the two-sided p-value is twice the smaller share of resamples in which the difference is at least zero or at most zero. The method treats recordings as independent, although several recordings come from the same speaker, so the intervals may be somewhat narrow.

4.6 Experimental Setup and Reproducibility

All experiments were run in Google Colab notebooks that read from and write to a shared Google Drive folder; each step resumes from its saved checkpoints. Table 4.3 lists the notebooks.

Table 4.3: Notebooks used in this work.

Notebook	Role
Sylheti_Data_Cleaning	build metadata and apply corrections
Step1b_Recording_Check, Step1c_Recheck	automatic recording check
Step2d_ASR_Turbo	fine-tune and test Whisper-large-v3-turbo
Step2e_ASR_Small	fine-tune and test Whisper-small
Step3_MT_Training	fine-tune and test NLLB
Step3b_Cascade	cascade ASR → MT evaluation
Step4_TTS_normalized, Step4b_TTS_Compare	TTS generation and evaluation
Step8_Round2_Organize	organise and convert Round 2 recordings
Step9_Round2_Recheck	Round 2 recording check
Step10_ASR_Round2	Round 2 training, test and bootstrap, Whisper-large-v3-turbo
Step10b_ASR_Round2_Small	Round 2 training, test and bootstrap, Whisper-small
Step11_Cascade_Round2	Round 2 cascade ASR → MT evaluation and bootstrap, all four ASR models
Step7_MOS_prepare	audio samples for the listening test (MOS)
Step7b_MOS_analysis	listening-test (MOS) results and significance tests
Step12_Demo_Round2	interactive end-to-end demonstration (Gradio)
Step12_Demo_Round2_V1voice	demonstration with the optional V1 voice
Step6_TTS_VoiceClone	voice cloning, attempt 1 (whole model fine-tuned)
Step6b_TTS_VoiceClone_v2	voice cloning, attempt 2 (stricter data filter, lower learning rate)
Step6c_TTS_VoiceClone_v3	voice cloning, attempt 3 (text encoder and duration predictor frozen)

[VERIFY: training time per model, if recorded.]

Chapter 5: Experiments and Results

Fine-tuning makes Sylheti ASR usable, Whisper-large-v3-turbo outperforms Whisper-small for every speaker and test set, the fine-tuned MT model reaches 51.32 BLEU (Sylheti → Bangla) on gold text, and the better ASR model also gives the better translation in every cascade setting. Sections 5.1–5.7 report the results of each module and of the cascade with the Round 1 data, and Section 5.8 reports how adding the Round 2 speakers changes ASR accuracy and translation quality.

5.1 ASR Training Behaviour



Figure 5.1: Validation WER (left) and training and validation loss (right) for Whisper-large-v3-turbo and Whisper-small. Circles mark the selected checkpoints.

Table 5.1: Validation WER / CER (%) during fine-tuning (300 validation clips). Selected checkpoints in bold.

Step	Turbo WER / CER	Small WER / CER
250	53.28 / 20.64	63.73 / 25.37
500	46.32 / 18.04	53.38 / 20.90
750	42.99 / 17.12	50.00 / 19.48
1,000	40.20 / 15.46	46.77 / 17.98
1,250	44.53 / 19.50	45.97 / 17.80
1,500	41.04 / 16.88	44.38 / 17.88
1,750	39.65 / 16.12	43.38 / 17.12
2,000	40.90 / 18.20	42.59 / 16.77
2,250	**35.92 / 14.14**	42.49 / 16.98
2,500	36.27 / 14.17	44.63 / 17.71
2,750	35.97 / 14.27	43.68 / 16.83

Step	Turbo WER / CER	Small WER / CER
3,000	36.22 / 14.21	**41.84 / 17.06**
3,250	36.02 / 14.01	42.19 / 17.13

Whisper-large-v3-turbo has a lower validation WER than Whisper-small at every evaluation step. Its best checkpoint is step 2,250 (WER 35.92%); it was last evaluated at step 3,250, and training ended at step 3,265, the end of the fifth epoch. Whisper-small reached its best WER at step 3,000 (41.84%) and completed all 3,270 steps. Training loss keeps falling for both models while validation loss flattens after about 1,500 steps, which indicates the start of overfitting.

5.2 ASR Test Results

Table 5.2: ASR test results, WER / CER (%). Best results in bold.

Model	test_seen (2,597 recordings)	test_unseen, A2 (928 recordings)
Whisper-large-v3-turbo, zero-shot	138.6 / 105.9	130.6 / 91.9
Whisper-small, fine-tuned	43.1 / 15.4	42.5 / 12.8
Whisper-large-v3-turbo, fine-tuned	**37.3 / 13.0**	**33.9 / 9.2**

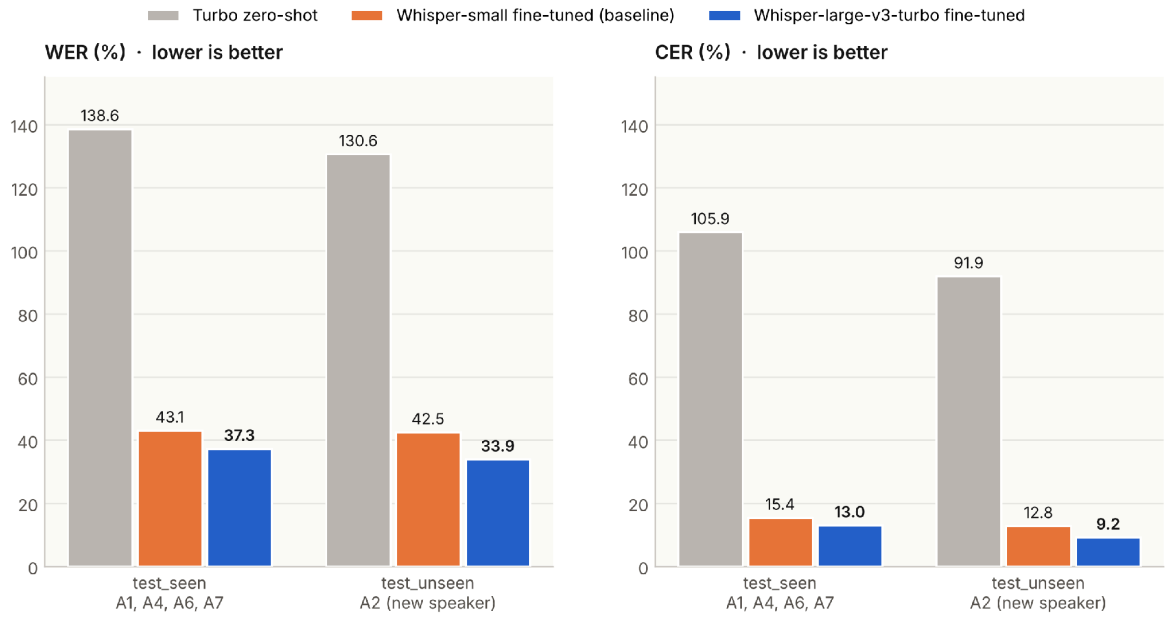


Figure 5.2: ASR test WER and CER for zero-shot Whisper-large-v3-turbo, fine-tuned Whisper-small and fine-tuned Whisper-large-v3-turbo.

Without fine-tuning, Whisper cannot transcribe Sylheti: its WER exceeds 100% on both test sets. Fine-tuning reduces WER to 37.3% (seen) and 33.9% (unseen) for turbo, and to

43.1% and 42.5% for Whisper-small. CER is much lower than WER for both models (9.2–15.4%), meaning that many word errors are near misses at the character level.

5.3 Per-Speaker Results

Table 5.3: Per-speaker test WER / CER (%).

Speaker (test set)	Turbo zero-shot	Whisper-small	Turbo fine-tuned
A1 (seen)	117.5 / 107.6	40.7 / 12.8	31.5 / 9.2
A4 (seen)	167.3 / 138.2	54.7 / 22.7	47.7 / 19.6
A6 (seen)	116.9 / 82.2	41.2 / 14.2	37.0 / 12.4
A7 (seen)	133.9 / 96.9	32.2 / 8.7	**26.2 / 6.4**
A2 (unseen)	130.6 / 91.9	42.5 / 12.8	33.9 / 9.2

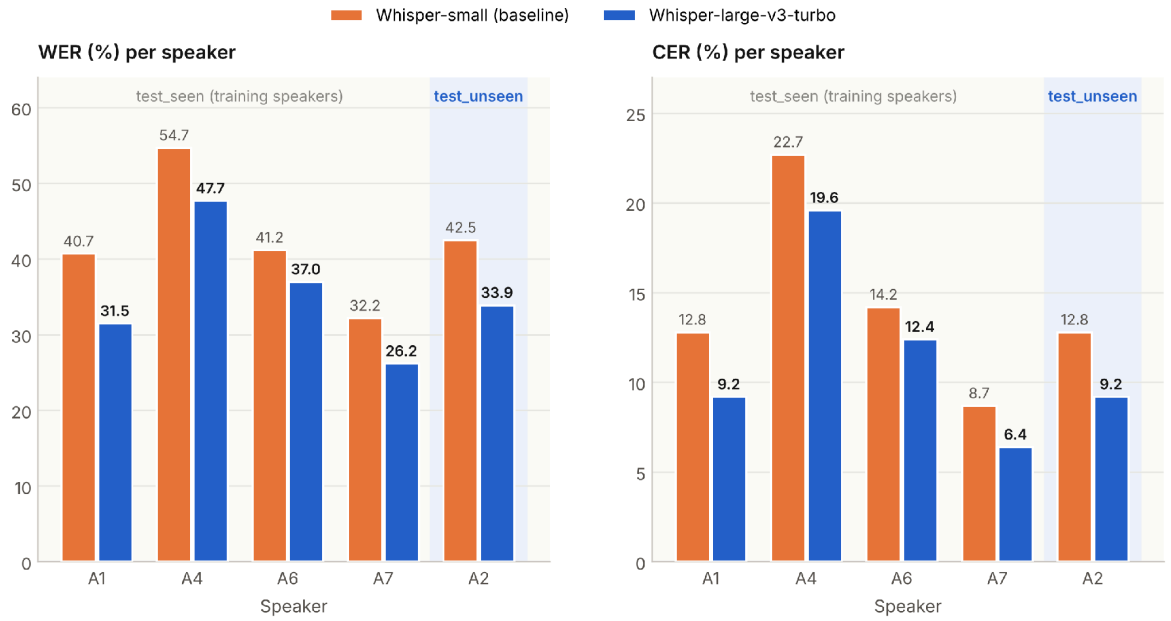


Figure 5.3: Per-speaker test WER and CER for the two fine-tuned models.

A4 is the hardest speaker for both fine-tuned models (turbo WER 47.7%), and A7 the easiest (26.2%). With turbo, the unseen speaker A2 (33.9%) is recognised better than two of the four training speakers (A4 and A6), so the fine-tuned model is not limited to the voices it was trained on.

5.4 MT Results

Table 5.4: MT results on 932 test sentences with gold Sylheti input, BLEU / chrF.

System	Sylheti → Bangla	Sylheti → English
Copy source (no translation)	12.58 / 49.56	–
NLLB zero-shot	7.89 / 38.45	10.28 / 33.65
NLLB fine-tuned	**51.32 / 76.71**	**38.25 / 59.11**

Table 5.5: MT training and validation loss (best checkpoint in bold).

Step	250	500	750	1,000	1,250	1,500	1,750	2,000
Train loss	1.2548	0.8942	0.7829	0.5770	0.5730	0.4262	0.4466	0.3331
Validation loss	1.1606	0.9986	0.9501	0.9536	**0.9284**	0.9634	0.9590	1.0103

Zero-shot NLLB scores below the copy-source baseline for Bangla (7.89 vs 12.58 BLEU): simply copying the Sylheti sentence overlaps more with the Bangla reference than the untrained model's output does. Fine-tuning raises BLEU to 51.32 for Bangla and 38.25 for English. The best checkpoint is step 1,250; validation loss rises afterwards while training loss keeps falling, and training stopped at step 2,000.

5.5 Cascade ASR → MT Results

Table 5.6: Cascade results, BLEU / chrF. Best ASR-based results in bold.

Input to MT	seen → Bangla	seen → English	unseen → Bangla	unseen → English
Whisper-small output	30.37 / 61.59	26.08 / 47.18	30.91 / 63.27	27.17 / 48.22
Whisper-large-v3-turbo output	**34.21 / 65.62**	**29.63 / 50.91**	**37.71 / 68.86**	**31.30 / 52.50**
Gold Sylheti text (upper bound)	51.25 / 76.61	38.20 / 59.02	51.28 / 76.70	38.20 / 59.06

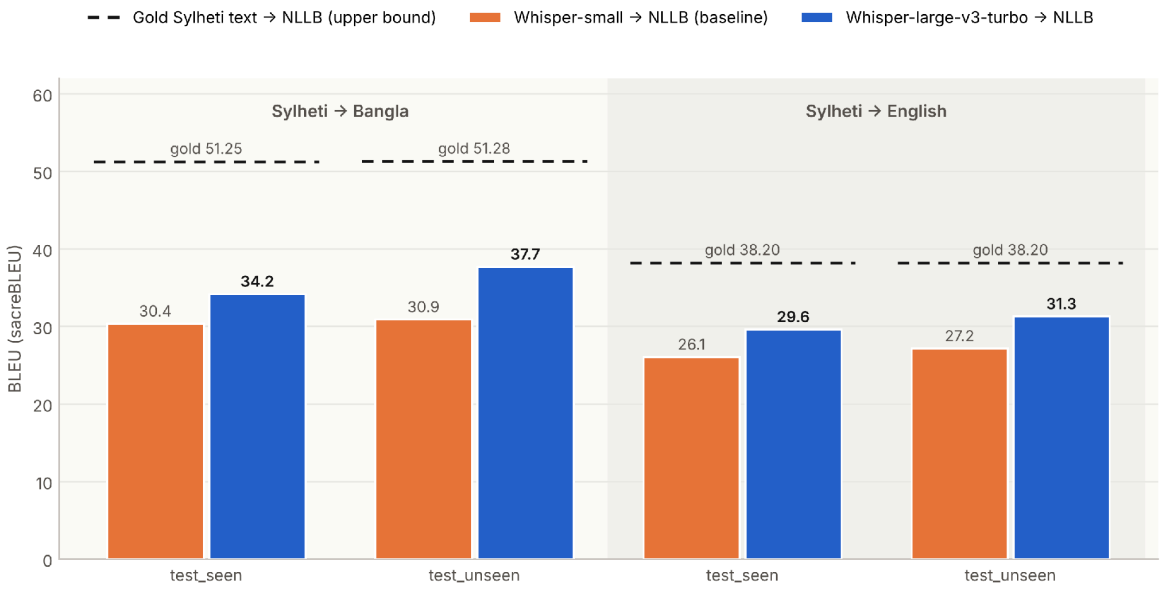


Figure 5.4: Cascade BLEU for Whisper-small and Whisper-large-v3-turbo input, with the gold-text upper bound (dashed).

The better ASR model gives the better translation in every direction and test set. ASR errors cost 14–17 BLEU (Bangla) and 7–9 BLEU (English) with turbo, and 20–21 BLEU (Bangla) and 11–12 BLEU (English) with Whisper-small. The gold-text scores are almost identical for the two test sets, because both are drawn from the same MT test sentences (932 and 928 sentences). No test sentence appears in the MT training data.

5.6 TTS Results

Table 5.7: TTS on 100 MT translations, back-transcribed with Whisper-large-v3-turbo. "Mapped" = Devanagari letters in the transcript mapped to Bengali.

System and setting	Language	CER / WER (raw)	CER / WER (mapped)	Audio (min)	RTF
MMS-TTS, raw text, first run	Bangla	52.5 / 95.2	–	6.8	0.043
MMS-TTS, normalised text, 200 tokens	Bangla	54.0 / 96.3	39.0 / 91.4	6.8	0.037
MMS-TTS, normalised text, 440 tokens	Bangla	60.4 / 104.5	45.4 / 99.6	6.8	0.049
MMS-TTS	English	5.6 / 11.3	–	5.6	0.038

English TTS is clearly intelligible (CER 5.6%). Bangla appears far worse, but much of this gap comes from the evaluator. Even with the language forced to Bengali, Whisper wrote Devanagari (Hindi) letters in 85 of the 100 Bangla clips; mapping these letters to Bengali lowers the MMS CER by about 15 points (54.0% → 39.0% with a 200-token limit). Raising the token limit from 200 to 440 made all Bangla scores worse (MMS CER 60.4%, WER 104.5%): with more room, Whisper inserts repeated or invented words, which the shorter limit had cut off. The scores therefore depend strongly on decoding settings.

Text normalisation did not improve the scores. A human listening test (MOS) is planned as the main measure of Bangla TTS quality. [VERIFY: MOS results after the listening test.] MMS-TTS generates speech much faster than real time (RTF 0.037–0.049), so it is suitable for live use.

5.6.1 Voice Cloning (Additional Experiment)

To test whether the system can speak its output in the voice of a Sylheti speaker, the Bangla MMS-TTS model was fine-tuned on about 1.9 hours of one speaker's recordings (Section 4.4.1). Table 5.8 lists the three training settings and Table 5.9 the intelligibility of the generated speech on 50 held-out sentences.

Table 5.8: Voice-cloning training settings (speaker V1).

Setting	Attempt 1	Attempt 2	Attempt 3
Training data (clip CER filter)	1,529 clips (≤ 50%)	about 1,440 clips (≤ 25%)	1,529 clips (≤ 50%)
Trained parts	whole model	whole model	posterior encoder, flow, decoder (text encoder and duration predictor frozen)
Learning rate	2e-5	1e-5	2e-5
Steps (batch 16)	5,000	2,000	5,000
Precision	fp16	fp32	fp32
Checkpoints evaluated	final	every 500 steps	every 1,000 steps

Table 5.9: Voice-cloning results: CER (%) of the generated speech on 50 held-out sentences, back-transcribed with the fine-tuned Whisper-large-v3-turbo model (lower is better).

System	Default noise	noise_scale 0.3
Real recordings of the same sentences	11.9	–
MMS-TTS, original voice (no fine-tuning)	20.2	**16.8**
Attempt 1, step 5,000	64.2	49.7
Attempt 2, best (step 2,000)	75.5	61.0
Attempt 3, step 1,000	31.4	29.2
Attempt 3, step 2,000	33.9	**28.1**
Attempt 3, step 3,000	34.7	28.8
Attempt 3, step 4,000	36.1	29.7

Attempt 3, step 5,000	60.7	30.0

Fine-tuning the whole model (attempts 1 and 2) made the speech much less intelligible than the original MMS voice (best CER 49.7% vs 16.8%), although the training recordings are clean (median clip CER 4.4% for the Round 1 recordings and 8.2% for the web-recorder recordings). Keeping the text encoder and duration predictor fixed (attempt 3) halved the error with the same data, learning rate and number of steps: the best V1 model reaches 28.1% CER (step 2,000, noise_scale 0.3). Most of this gain is reached after 1,000 steps. With the default inference noise, longer training makes the output unstable (CER 31.4% at step 1,000, 60.7% at step 5,000), whereas with noise_scale 0.3 the CER stays near 30%; lower inference noise improved every model.

The adapted voice remains about 11 points less intelligible than the original MMS voice, and informal listening revealed a buzzing artefact in the adapted voice. MMS-TTS with its original voice therefore remains the default Bangla TTS of the pipeline; the V1 model is reported as an additional experiment. Voice similarity and naturalness have not yet been measured and require a listening test. [VERIFY: MOS / voice-similarity results, if a listening test is carried out.]

5.7 End-to-End Demonstration

An interactive demonstration of the full pipeline was built with Gradio (Step12_Demo_Round2, Table 4.3) and runs in Google Colab. Input can be Sylheti speech recorded with a microphone, an uploaded audio file in common phone formats, or typed Sylheti text. The interface shows the recognised Sylheti text and the Bangla and English translations, and plays the synthesised Bangla and English speech (Figure 5.5). The translations and the recognised text can be corrected by the user and synthesised again. The demonstration uses the Round 2 Whisper-large-v3-turbo model (Section 5.8), the fine-tuned NLLB model and MMS-TTS, with the same decoding settings and Bangla text normalisation as in the evaluation. As an option, the Bangla translation and the Sylheti text can also be spoken in the voice of speaker V1 with the adapted MMS-TTS model of Section 5.6.1.

On a Colab L4 GPU, processing one utterance from speech input to the Bangla and English speech outputs took 2.1 s on average over five Sylheti voice messages recorded on a phone (range 0.8–3.9 s): 0.9 s for speech recognition, 0.9 s for translation into both languages and 0.3 s for speech synthesis; the optional V1 voice added 0.2 s. Longer messages took longer, mainly in recognition and translation. These messages were spontaneous rather than read speech. In an informal check, the short messages were mostly translated correctly, while the longest message contained clear recognition and translation errors, in line with the limitation in Section 6.6.

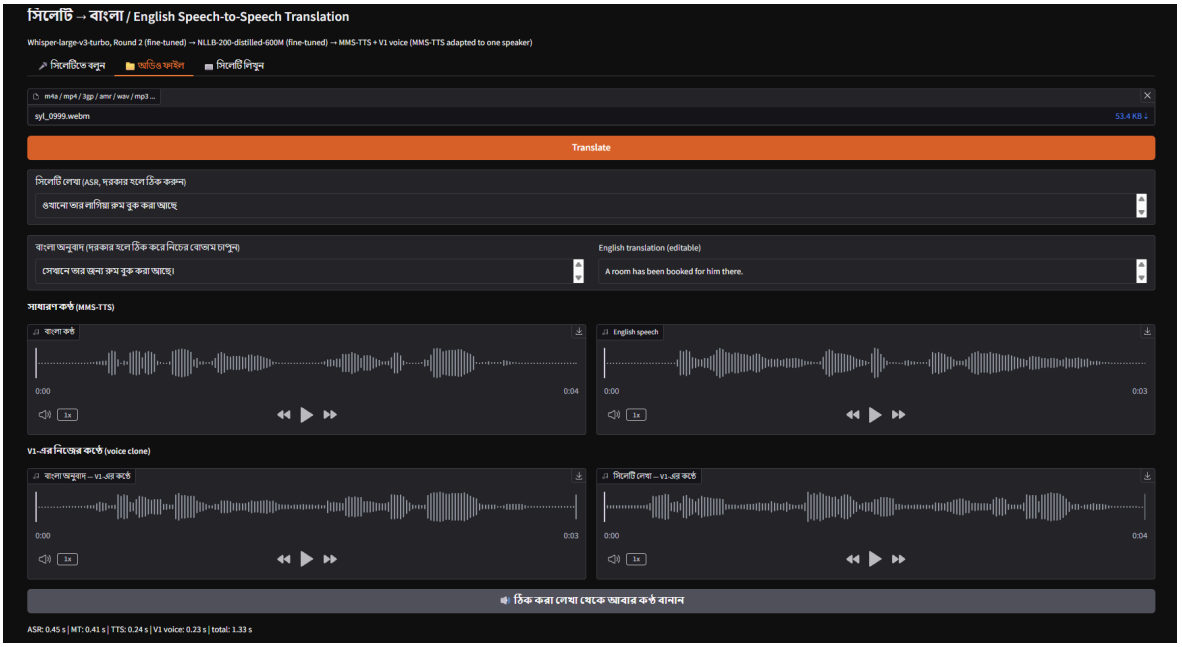


Figure 5.5: Interactive end-to-end demonstration: Sylheti speech input, recognised Sylheti text, Bangla and English translations, and synthesised speech in the standard MMS-TTS voice and the adapted V1 voice. The recognised text and translations can be edited and re-synthesised.

5.8 Effect of Additional Speakers (Round 2)

To answer RQ4, Whisper-large-v3-turbo was trained again from the public pretrained checkpoint on the Round 1 training data together with the Round 2 training data (Section 3.7, Table 3.6), with the settings of Table 4.1. In this section, the model trained on Round 1 data only (Sections 5.1–5.3) is called the Round 1 model and the new model the Round 2 model. Whisper-small was retrained in the same way (Section 5.8.4), and the cascade was re-run with the ASR output of all four models (Section 5.8.5).

5.8.1 Training Behaviour

Table 5.10: Validation WER / CER (%) of Whisper-large-v3-turbo trained on Round 1 data and on Round 1 + Round 2 data (same 300 validation clips). Selected checkpoints in bold.

Step	Round 1 data	Round 1 + Round 2 data
250	53.28 / 20.64	52.24 / 19.54
500	46.32 / 18.04	46.02 / 17.95
750	42.99 / 17.12	43.08 / 16.45
1,000	40.20 / 15.46	43.93 / 18.98
1,250	44.53 / 19.50	45.52 / 19.16
1,500	41.04 / 16.88	39.65 / 15.87
1,750	39.65 / 16.12	37.06 / 14.89
2,000	40.90 / 18.20	38.01 / 15.68
2,250	**35.92 / 14.14**	36.42 / 14.60
2,500	36.27 / 14.17	**35.52 / 14.10**
2,750	35.97 / 14.27	37.31 / 14.20
3,000	36.22 / 14.21	36.17 / 13.82
3,250	36.02 / 14.01	36.32 / 14.27
3,500	–	36.02 / 14.23
3,750	–	35.97 / 14.34

The Round 2 model reached its lowest validation WER at step 2,500 (35.52%, CER 14.10%), slightly below the Round 1 model (35.92%, CER 14.14% at step 2,250). As in Round 1, WER rose temporarily at steps 1,000–1,250 while the validation loss kept falling. Training was ended by the early-stopping rule after more than 1,000 steps without a new best WER (last evaluation at step 3,750), and the checkpoint from step 2,500 was kept. With 28,301 training recordings, five epochs would correspond to 4,420 steps.

5.8.2 Test Results

Table 5.11: Test WER / CER (%) of the Round 1 and Round 2 Whisper-large-v3-turbo models, and the difference (Round 2 − Round 1) with 95% confidence intervals and p-values from a paired bootstrap (10,000 resamples). Differences are computed from unrounded values.

Test set (recordings)	Round 1 model	Round 2 model	WER difference [95% CI]	CER difference [95% CI]
test_seen (2,597)	37.3 / 13.0	**36.1 / 12.3**	−1.2 [−2.4, −0.1], p = 0.035	−0.7 [−1.6, 0.1], p = 0.096
test_unseen, A2 (928)	34.0 / 9.2	**31.5 / 8.3**	−2.6 [−3.6, −1.5], p < 0.0001	−0.9 [−1.3, −0.6], p < 0.0001
test_new (128)	41.0 / 11.5	**36.8 / 10.1**	−4.1 [−6.8, −1.6], p = 0.001	−1.4 [−2.4, −0.5], p = 0.002
test_new_all (1,449)	32.6 / 9.2	**28.9 / 7.8**	−3.6 [−4.5, −2.7], p < 0.0001	−1.4 [−1.7, −1.1], p < 0.0001
test_r2_seen (895)	41.9 / 12.5	**35.1 / 9.7**	−6.8 [−7.9, −5.7], p < 0.0001	−2.8 [−3.2, −2.4], p < 0.0001

The Round 2 model has a lower WER and CER than the Round 1 model on every test set. The improvement is largest for speakers whose voices the Round 1 model had not heard: WER fell by 4.1 points on the three new speakers (41.0% to 36.8%), by 2.6 points on A2 and by 6.8 points on the test sentences of the Round 2 training speakers. For the Round 1 training speakers (test_seen) the gain is small: WER fell by 1.2 points ($p = 0.035$), and the CER difference is not significant, as its 95% confidence interval includes zero ($p = 0.096$).

Significance was tested with the paired bootstrap of Section 4.5.5. With a Bonferroni correction for the ten comparisons ($\alpha = 0.005$), all differences remain significant except the two on test_seen. test_new contains only test sentences and is the main measure for new speakers; test_new_all also contains training sentences, which the Round 2 model has heard from other speakers, and is reported as additional information. The Round 1 model was re-evaluated in the same run; its scores match Table 5.2 except for A2 (WER 34.0% here, 33.9% in Table 5.2).

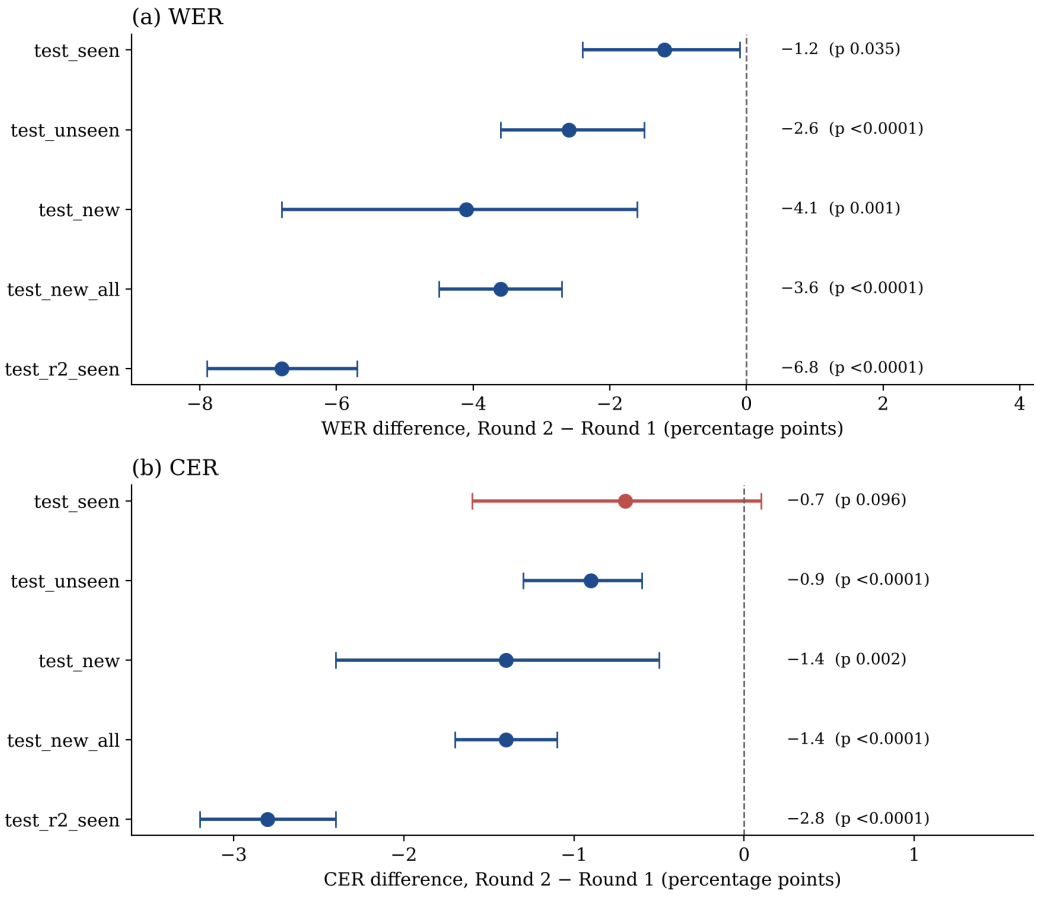


Figure 5.6: Paired bootstrap comparison of the Round 2 and Round 1 models: difference in WER (a) and CER (b), Round 2 − Round 1, with 95% confidence intervals. Negative values favour the Round 2 model; red marks the interval that includes zero.

5.8.3 Per-Speaker Results

Table 5.12: Per-speaker test WER / CER (%) of the Round 2 Whisper-large-v3-turbo and Whisper-small models.

Speaker	Group	Test recordings	Whisper-large-v3 -turbo WER / CER	Whisper-small WER / CER
A1	Round 1 training	91	31.0 / 9.2	37.9 / 11.9
A4	Round 1 training	838	45.7 / 18.7	51.5 / 21.9
A6	Round 1 training	903	35.5 / 11.3	39.4 / 13.5
A7	Round 1 training	765	26.4 / 6.5	30.1 / 8.0
A2	unseen	928	31.5 / 8.3	37.6 / 10.8
B01	Round 2 training	94	37.3 / 10.8	43.7 / 14.6

B03	Round 2 training	109	34.1 / 7.9	40.6 / 11.0
B05	Round 2 training	131	36.3 / 10.4	42.8 / 13.7
B06	Round 2 training	19	38.6 / 10.3	45.0 / 14.5
B07	Round 2 training	119	36.0 / 11.0	41.7 / 13.9
B08	Round 2 training	184	28.7 / 6.9	35.2 / 9.6
B10	Round 2 training	118	37.3 / 11.6	46.1 / 18.7
B11	Round 2 training	120	38.9 / 10.5	50.1 / 16.2
B12	Round 2 training	1	not reported (1 recording)	not reported (1 recording)
B02	new unseen	35	40.7 / 11.2	41.5 / 13.7
B04	new unseen	53	31.4 / 7.4	35.5 / 10.2
B09	new unseen	40	39.6 / 12.2	49.7 / 16.9

Compared with the Round 1 model (Table 5.3), WER fell for A1 (31.5% to 31.0%), A4 (47.7% to 45.7%), A6 (37.0% to 35.5%) and A2 (33.9% to 31.5%), and stayed almost the same for A7 (26.2% to 26.4%). A4 remains the hardest speaker. The new speakers B02, B04 and B09 reach 31.4–40.7% WER, within the range of the training speakers, but each has only 35–53 test recordings, and B12 has a single test recording, so the per-speaker values of these speakers are uncertain.

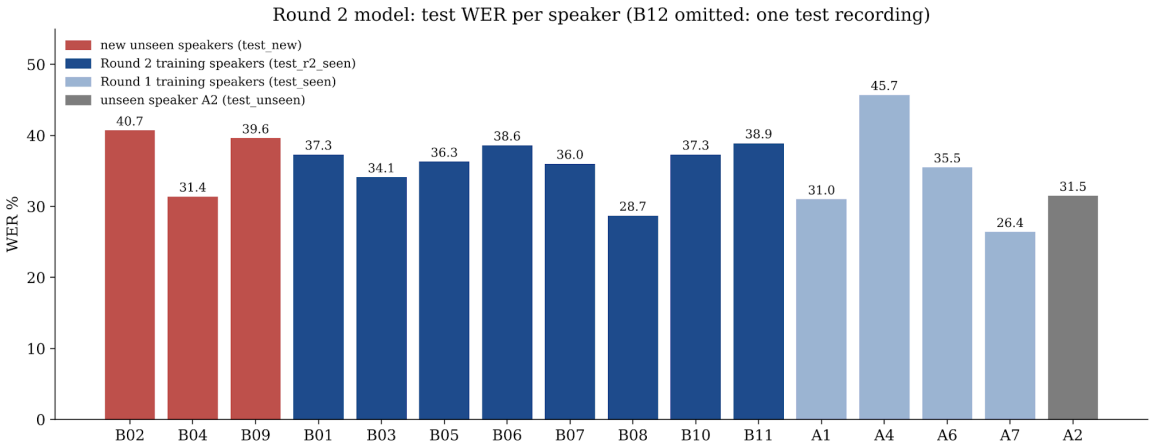


Figure 5.7: Per-speaker test WER of the Round 2 model. New speakers are scored on test_new; B12 is omitted because it has one test recording.

5.8.4 Whisper-small

Whisper-small was trained again in the same way, from the public pretrained checkpoint on the Round 1 + Round 2 training data with the settings of Table 4.1, on an A100 GPU. Its lowest validation WER was 40.60% (CER 16.23%) at step 2,500, compared with 41.84% at step 3,000 for the Round 1 Whisper-small model (Table 5.13). Training stopped at step 3,500, after 1,000 steps without a new best validation WER. Table 5.14 compares the two Whisper-small models on the same test recordings.

Table 5.13: Validation WER / CER (%) of Whisper-small trained on Round 1 data and on Round 1 + Round 2 data (same 300 validation clips). Selected checkpoints in bold.

Step	Round 1 data	Round 1 + Round 2 data
250	63.73 / 25.37	64.18 / 25.57
500	53.38 / 20.90	54.58 / 22.51
750	50.00 / 19.48	49.05 / 19.74
1,000	46.77 / 17.98	46.52 / 18.62
1,250	45.97 / 17.80	46.22 / 18.01
1,500	44.38 / 17.88	43.38 / 17.32
1,750	43.38 / 17.12	43.28 / 16.92
2,000	42.59 / 16.77	42.24 / 16.35
2,250	42.49 / 16.98	41.54 / 16.31
2,500	44.63 / 17.71	**40.60 / 16.23**
2,750	43.68 / 16.83	41.24 / 16.54
3,000	**41.84 / 17.06**	41.49 / 16.52
3,250	42.19 / 17.13	41.79 / 16.87
3,500	–	40.95 / 16.24

Table 5.14: Test WER / CER (%) of the Round 1 and Round 2 Whisper-small models, and the difference (Round 2 − Round 1) with 95% confidence intervals and p-values from a paired bootstrap (10,000 resamples). Differences are computed from unrounded values.

Test set (recordings)	Round 1 model	Round 2 model	WER difference [95% CI]	CER difference [95% CI]

test_seen (2,597)	43.2 / 15.7	**40.7 / 14.7**	−2.5 [−3.4, −1.6], p < 0.0001	−1.0 [−1.6, −0.5], p = 0.0008
test_unseen, A2 (928)	42.5 / 12.9	**37.6 / 10.8**	−4.9 [−6.0, −3.9], p < 0.0001	−2.1 [−2.5, −1.7], p < 0.0001
test_new (128)	48.3 / 16.2	**41.9 / 13.4**	−6.5 [−9.4, −3.7], p < 0.0001	−2.7 [−3.9, −1.6], p < 0.0001
test_new_all (1,449)	41.7 / 13.4	**36.5 / 11.1**	−5.3 [−7.2, −3.8], p < 0.0001	−2.4 [−3.1, −1.7], p < 0.0001
test_r2_seen (895)	52.1 / 17.9	**42.4 / 13.6**	−9.7 [−10.9, −8.5], p < 0.0001	−4.3 [−5.0, −3.5], p < 0.0001

All ten differences are significant, also after a Bonferroni correction ($\alpha = 0.005$), including the CER difference on test_seen, which was not significant for Whisper-large-v3-turbo. Whisper-small gains more from the additional speakers than the larger model: its WER fell by 6.5 points on the new speakers (Whisper-large-v3-turbo: 4.1), by 4.9 points on A2 (2.6), by 9.7 points on the Round 2 training speakers (6.8) and by 2.5 points on the Round 1 training speakers (1.2). The Round 2 Whisper-small model is nevertheless less accurate than the Round 1 Whisper-large-v3-turbo model on every test set (for example 41.9% vs 41.0% WER on test_new and 37.6% vs 34.0% on A2), so additional speakers do not replace a larger model. As with Whisper-large-v3-turbo, A4 remains the hardest Round 1 speaker (51.5% WER) and B09 the hardest new speaker (49.7%). The Round 1 Whisper-small model was re-evaluated in the same run; its scores on test_seen and test_unseen (43.2 / 15.7 and 42.5 / 12.9) differ slightly from Table 5.2 (43.1 / 15.4 and 42.5 / 12.8), and the values from the same run are used for the comparison.

5.8.5 Effect on the Cascade

To test whether the better ASR models also give better translations, the ASR output of all four models (Whisper-large-v3-turbo and Whisper-small, each trained on Round 1 data and on Round 1 + Round 2 data) was translated with the unchanged NLLB model, using the settings of Section 5.5. Only test sets whose sentences belong to the MT test split are used; test_new_all is excluded because most of its sentences were used to train the MT model. The gold-text upper bound on the same recordings is 51.25 / 38.20 BLEU (Bangla / English) on test_seen, 51.28 / 38.20 on test_unseen, 49.40 / 40.21 on test_new and 50.10 / 35.20 on test_r2_seen; for test_seen and test_unseen it is identical to Table 5.6. Differences

were tested with a paired bootstrap over recordings on corpus-level BLEU and chrF (10,000 resamples).

Table 5.15: Cascade results with the Round 1 and Round 2 ASR models, Round 1 → Round 2; the gold Sylheti text rows give the upper bound without ASR errors. * The difference is significant in a paired bootstrap test ($p < 0.05$).

Test set (recordings)	ASR model	Sylheti → Bangla BLEU	Sylheti → Bangla chrF	Sylheti → English BLEU	Sylheti → English chrF
test_seen (2,597)	Whisper-large-v3-turbo	34.45 → 35.83*	65.62 → 66.06	29.58 → 30.26	50.81 → 51.38*
	Whisper-small	30.66 → 31.49*	61.71 → 62.76*	26.12 → 27.25*	47.12 → 48.08*
	Gold Sylheti text (upper bound)	51.25	76.61	38.20	59.02
test_unseen, A2 (928)	Whisper-large-v3-turbo	37.39 → 39.46*	68.56 → 69.80*	31.63 → 32.81*	52.71 → 54.01*
	Whisper-small	30.83 → 33.93*	63.29 → 66.25*	27.10 → 28.71*	48.40 → 50.18*
	Gold Sylheti text (upper bound)	51.28	76.70	38.20	59.06
test_new (128)	Whisper-large-v3-turbo	35.41 → 35.91	67.58 → 68.75	31.53 → 33.18	51.64 → 53.21
	Whisper-small	27.38 → 29.62	60.18 → 63.19*	26.21 → 27.85	45.52 → 47.32
	Gold Sylheti text (upper bound)	49.40	77.65	40.21	59.23
test_r2_seen (895)	Whisper-large-v3-turbo	29.86 → 34.79*	62.89 → 66.57*	25.07 → 27.38*	47.60 → 50.03*
	Whisper-small	22.41 → 29.00*	55.76 → 61.63*	20.32 → 23.77*	42.73 → 46.12*
	Gold Sylheti text (upper bound)	50.10	75.75	35.20	57.41

Translation quality is higher with the Round 2 ASR models in all 32 comparisons. The gain is significant in every comparison on A2 and on the Round 2 training speakers, where Sylheti → Bangla BLEU rose by 2.1 and 4.9 points with Whisper-large-v3-turbo and by

3.1 and 6.6 points with Whisper-small. On test_seen the gain is about one BLEU point; it is significant in all four comparisons for Whisper-small but in only two for Whisper-large-v3-turbo. On the new speakers only one of eight comparisons is significant (Whisper-small, Sylheti → Bangla chrF, p = 0.008), although the ASR gains on the same recordings are significant. With 128 recordings the confidence intervals are wide (for example −2.74 to 3.92 BLEU for Whisper-large-v3-turbo, Sylheti → Bangla), so a translation gain for new speakers can neither be confirmed nor ruled out. Even the best system (Whisper-large-v3-turbo, Round 2) remains 15.4 BLEU (Bangla) and 7.9 BLEU (English) below the gold-text upper bound on test_seen. Because the Round 1 ASR models were re-evaluated, their cascade scores differ slightly from Table 5.6 (for example 34.45 instead of 34.21 BLEU for Whisper-large-v3-turbo on test_seen).

Table 5.16 shows five random test_new recordings translated by the Round 2 Whisper-large-v3-turbo cascade.

Table 5.16: Example cascade outputs (Whisper-large-v3-turbo, Round 2; five random test_new recordings). The ASR column shows the key words that were misrecognised.

#	ASR error (key words)	English reference	English cascade output	Source of the error
1	অইলো দুক্কুর → অইল দুপুর	The time is two in the afternoon.	It's two in the afternoon.	none (translation correct)
2	উঠা → উঁটা	Have to get on a plane.	I'll have to go by plane.	small ASR error; meaning kept
3	শরীফ → শরিফ (spelling only)	The holy shrine of the martyred intellectuals.	The Sharif of the Martyrs' Quarters.	MT
4	রোগা → রুগা	The fatter the sahib, the skinnier the young woman on his arm.	The fatter a gentleman is, the more ugly his bride is.	ASR error changes the meaning
5	ফাইলাম → ভাইলাম	I found a group where there's no crying going on.	A group of brothers where there's no crying.	ASR error changes the meaning

In examples 1 and 2 the MT model recovers the meaning from an imperfect transcript. In examples 4 and 5 a single misrecognised word changes the meaning of the translation: রোগা (thin) recognised as রুগা is translated as “ugly”, and ফাইলাম (found) recognised as

ভাইলাম leads to “a group of brothers”. In example 3 the transcript differs only in spelling, but the MT model translates the expression মাজার শরীফ (holy shrine) as a proper name.

Chapter 6: Discussion

The results support four main conclusions: a larger pretrained ASR model adapts better to Sylheti with the same data, recognition quality depends strongly on the individual speaker, ASR errors are the largest source of quality loss in the cascade, and adding speakers mainly improves recognition of speakers not seen in training. This chapter interprets these findings and states the limitations of the study.

6.1 Large vs Small ASR Model

With identical data and split, Whisper-large-v3-turbo reaches 37.3% WER on test_seen and 33.9% on test_unseen, against 43.1% and 42.5% for Whisper-small. The advantage is larger on the unseen speaker (8.6 points) than on the seen speakers (5.8 points), which suggests that the larger model's pretraining helps it generalise to a new voice. The gap remains in Round 2: Whisper-small gains more from the additional speakers than turbo (e.g. 4.9 vs 2.6 WER points on A2), but the Round 2 Whisper-small model is still worse than even the Round 1 turbo model on every test set (e.g. test_seen 40.7% vs 37.3% WER), so more speakers do not replace a larger model.

The comparison is not perfectly controlled. The two models were trained on different GPUs with different precision (bf16 vs fp16), and Whisper-small was resumed after step 2,250 with a restarted learning-rate schedule, so its best checkpoint (step 3,000) was trained at a much higher learning rate than turbo at the same step (8.5×10^{-6} vs 0.9×10^{-6}). However, turbo already had a lower validation WER than Whisper-small at every evaluation step before the restart (steps 250–2,250), so the main conclusion does not depend on this difference.

6.2 Speaker Variability

WER ranges from 26.2% (A7) to 47.7% (A4) for the same turbo model, a larger spread than the difference between the two models. Speaker A4 is also the speaker with by far the most corrections in the recording check (898 moved and 2,706 uncertain recordings, Table 3.1). Both observations point to A4's recordings being harder: possible causes include recording conditions, speaking style, or residual misalignments that the automatic check could not resolve. A systematic listening check of A4's recordings was not carried out, so the exact cause remains open (Section 6.6). Because A4 contributes 838 of the 2,597 test_seen recordings, it lowers the average test_seen score. In addition, the Round 1

recordings of sentences 5501–6000 filed under A4 were in fact read by speaker B01 (Section 4.4.1), so the A4 results partly include a second voice.

6.3 Generalisation to Unseen Speakers

Speaker A2 never appears in training, yet turbo recognises A2 better (33.9% WER) than two of the four training speakers. This is encouraging, but it rests on a single unseen speaker whose recordings are clean (8,061 of 9,318 recordings received an ok verdict). The result shows that generalisation is possible, not how well the model generalises to Sylheti speakers in general.

Round 2 tests this on three further speakers. The Round 1 model reached 41.0% WER on their test sentences, higher than on A2 (34.0%) and on three of the four training speakers, which suggests that A2 was an easier unseen speaker than a typical new one. Adding nine training speakers reduced WER by 4.1 points on the new speakers and by 2.6 points on A2, but by only 1.2 points on the Round 1 training speakers (Section 5.8.2). Greater speaker variety therefore mainly helps on voices the model has not heard. Part of this gain may also come from the recording conditions: all Round 2 speakers used the same web recorder, which the Round 2 model has seen during training and the Round 1 model has not, and the present experiment cannot separate the two effects.

The same speaker effect explains why the cascade scores higher on test_unseen than on test_seen (37.71 vs 34.21 BLEU for Bangla): the sentences are almost the same in both sets, as the nearly identical gold-text upper bounds show (51.25 and 51.28), but test_unseen contains only the clean speaker A2 while test_seen includes A4.

6.4 Error Propagation from ASR to Translation

With gold Sylheti input, the MT model reaches about 51 BLEU for Bangla and 38 for English. With ASR output, it loses 14–17 BLEU for Bangla and 7–9 BLEU for English using turbo, and 20–21 and 11–12 BLEU using Whisper-small. ASR is therefore the main bottleneck of the pipeline: the model with the lower ASR error also gave the higher BLEU in all four direction and test-set combinations, and improving ASR, for example with more speakers in Round 2, is the most direct way to improve the whole system. The smaller absolute loss for English is partly because its upper bound is lower. The Round 2 experiment supports this: with the Round 2 ASR models, BLEU and chrF rose in all 32 comparisons (Section 5.8.5), and the examples in Table 5.16 show that a single

misrecognised word can change the meaning of a translation, while some errors come from the MT model itself.

6.5 Evaluating Bangla TTS

Back-transcription with Whisper is a convenient, automatic measure of TTS intelligibility, and it works well for English (CER 5.6%). For Bangla it is unreliable for two reasons. First, Whisper mixes Devanagari letters into Bangla transcripts in more than 80% of clips even when the language is fixed. Second, it inserts repeated or invented words, so that WER exceeds 100% and the scores change with the decoding limit (MMS CER 54.0% with 200 tokens, 60.4% with 440). The reported Bangla scores therefore mix TTS errors with errors of the measuring model, and they do not show how intelligible the synthesised speech really is. A Bangla-specific ASR model as the evaluator and a human listening test (MOS) are needed before drawing firm conclusions.

The voice-cloning experiment (Section 5.6.1) adds a second observation. Adapting the whole MMS-TTS model to about 1.9 hours of one speaker changed the voice but damaged pronunciation, even though the training recordings were clean. Keeping the text encoder and duration predictor fixed preserved much of the original model's intelligibility (CER 28.1% instead of 49.7%), which indicates that, with this amount of data, only the parts of the model that shape the sound of the voice should be adapted. Since the fine-tuned Sylheti ASR model used as evaluator was trained partly on the same speaker, its scores for the V1 voice may be slightly optimistic; voice similarity and the buzzing artefact can only be judged by listeners.

6.6 Limitations

- **Read speech only.** All recordings are read sentences. In an informal live-microphone test with an earlier ASR model, spontaneous Sylheti speech was mostly misrecognised (roughly 80% errors by the authors' estimate; not a measured result). Performance on conversational speech is therefore expected to be much lower than reported here.
- **Few speakers.** Round 1 trains on four speakers, and the training data are unbalanced (A1 has 800 training recordings). Round 2 adds nine training and three further unseen speakers, but the new-speaker test set is small (128 recordings, 35–53 per speaker), and the two rounds were recorded with different tools and checked with different models (Section 3.7). For the same reason, the cascade gain for the new speakers could not be

shown to be significant (Section 5.8.5). In addition, the Round 1 recordings of sentences 5501–6000 filed under A4 were read by B01, so A4 is not a single speaker and B01's voice also appears in the Round 1 training and test_seen data.

- **No standard Sylheti spelling.**WER and CER count spelling variants of the same word as errors, so they overstate the true error rate. In addition, the English reference translations of the corpus were produced by machine translation [30], so English BLEU and chrF are measured against references that may themselves contain errors.
- **TTS evaluation.** Bangla TTS scores depend on an evaluator that is weak on Bangla (Section 6.5), and no human evaluation has been completed yet.
- **Training differences.** The learning-rate restart and different hardware of the Whisper-small run (Section 6.1).

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This thesis presented a cascaded speech-to-speech translation system from Sylheti to Standard Bangla and English, built on a self-recorded speech corpus of the 9,320 Sylheti sentences of the Kothon text corpus [30], with their Bangla and English translations, recorded in two rounds by 17 speakers (37,681 recordings, 43.72 hours). An automatic recording check based on Whisper transcription and fuzzy matching corrected 688 misplaced Round 1 recordings and restored 9,266 recordings that an earlier filter had removed. The research questions are answered as follows.

- **RQ1.** Untrained Whisper cannot transcribe Sylheti (WER above 100%). Fine-tuning Whisper-large-v3-turbo reduces WER to 37.3% on seen speakers and 33.9% on an unseen speaker, compared with 43.1% and 42.5% for Whisper-small trained on the same data. The larger model is better for every speaker, and its advantage on the unseen speaker (8.6 points) is larger than on the seen speakers as a whole (5.8 points).
- **RQ2.** The fine-tuned NLLB-200-distilled-600M model reaches 51.32 BLEU (Sylheti → Bangla) and 38.25 BLEU (Sylheti → English) on gold text. With ASR output from turbo, translation loses 14–17 BLEU for Bangla and 7–9 BLEU for English; with Whisper-small the loss is larger. ASR is therefore the main bottleneck of the pipeline.
- **RQ3.** English TTS is clearly intelligible by back-transcription (CER 5.6%). For Bangla, back-transcription with Whisper is unreliable: it writes Devanagari letters in more than 80% of clips and inserts words, and its scores change with the decoding limit. Even after mapping the Devanagari letters, MMS-TTS reaches only 45.4% CER and 99.6% WER by this measure, so a listening test is needed to judge Bangla TTS quality. [VERIFY: update after the MOS test.]
- **RQ4.** Adding nine speakers recorded with the web-based recorder (28,301 instead of 20,920 training recordings) improved Whisper-large-v3-turbo on every test set. The gain was largest for speakers not seen in training: WER fell from 41.0% to 36.8% on three new speakers and from 34.0% to 31.5% on A2, both significant in a paired bootstrap test ($p \leq 0.001$), while the Round 1 training speakers improved by only 1.2 points. Whisper-small gained even more from the additional speakers (WER −6.5 points on the new speakers and −4.9 on A2, all differences significant) but remained less accurate than Whisper-large-v3-turbo trained on Round 1 data alone. With the

Round 2 ASR models, translation quality rose on every test set; the gain was significant for A2 and the Round 2 speakers (up to +6.6 BLEU for Bangla), but too small to be confirmed on the 128 test recordings of the new speakers.

7.2 Future Work

- **More speakers.** Round 2 showed that additional speakers mainly improve recognition of new voices. Further collection with the web recorder should aim for more speakers, balanced by gender, age and sub-region, rather than more recordings per speaker.
- **Spontaneous speech.** Collect and evaluate conversational Sylheti, since the current corpus contains read speech only.
- **Spelling-tolerant evaluation.** Develop a spelling normalisation for Sylheti so that WER does not count spelling variants as errors.
- **Better Bangla TTS evaluation.** Use a Bangla-specific ASR model as the evaluator and complete a human listening test (MOS).
- **Deployment.** Turn the demonstration of Section 5.7 into a mobile or web application and test it with conversational speech and real users.
- **Direct models.** Once more data is available, compare the cascade with direct speech-to-speech translation models.
- **Voice cloning.** Remove the buzzing artefact of the adapted voice (denoised training audio, longer training of the decoder only), measure voice similarity and naturalness with a listening test, and clone the same voice for English, either zero-shot (e.g. XTTS-v2) or by fine-tuning MMS-TTS English on English recordings of the speaker.

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Appendices

Appendix A: Hyperparameters

The complete ASR and MT configurations are given in Tables 4.1 and 4.2. Settings not listed there were left at the Hugging Face Transformers defaults: weight decay 0, gradient clipping at norm 1.0, linear learning-rate schedule, training seed 42. The data split uses seed 13; the validation subset of 300 clips is sampled with seed 13.

Appendix B: Training Logs

Validation WER/CER for both ASR models (Table 5.1) and MT training and validation loss (Table 5.5) were taken from the saved trainer_state.json files (Whisper-large-v3-turbo checkpoint 3,265; Whisper-small checkpoint 3,270). Training loss was logged every 50 steps. [VERIFY: attach the full log tables if required by the department.]

Appendix C: Example Outputs

Example outputs of the full cascade (Whisper-large-v3-turbo, Round 2, five random test_new recordings) are given in Table 5.16 (Section 5.8.5), with the misrecognised key words, the English reference and the English cascade output, and the source of each error.

Appendix D: Notebooks and Data Folders

The Colab notebooks are listed in Table 4.3. Data, checkpoints and results are stored in a shared Google Drive folder. Recordings and speaker identities are not published. [VERIFY: whether code (without data) will be released, and where.]

Appendix E: Bangla Text Normalisation Rules

Applied before Bangla TTS (Section 4.4): digits to Bangla number words (including thousand, lakh, crore and decimals); %, & and similar symbols spelled out; quotation marks and brackets removed; "/" read as "অথবা"; dashes replaced by spaces; repeated punctuation collapsed; daṇḍa added at sentence end; sentences containing Latin letters flagged.